Predatory Trading

Replicating, verifying and extending Brunnermeier & Pedersen (2005)

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Abstract

Brunnermeier and Pedersen (2005) show that when a large trader is forced to liquidate and other large traders know it, those traders sell alongside it and buy back later, so the price overshoots and liquidity disappears exactly when it is needed. We implement the model as an exact discrete-time simulation and reproduce its propositions and figures. We then check the equilibria independently: a linear-programming best-response solver finds no profitable deviation from the paper's strategies, and traders starting from a naive strategy learn the predatory equilibrium by fictitious play. Finally we use the same tools for three extensions: collusion among predators, contagion with numerically computed survival hurdles, and predators with unequal positions. The code, an interactive app and 115 automated tests are available at `https://github.com/dudegladiator/predatory-trading`.

1 Introduction

Large traders fear forced liquidation, especially when their need to sell is known to others. Brunnermeier and Pedersen [1] (henceforth BP) model this situation and call the response *predatory trading*: trading that induces or exploits another investor's need to reduce a position. Their motivating examples include the collapse of Long-Term Capital Management in 1998, when counterparties allegedly sold the assets LTCM had to sell, and the 1987 crash, when aggressive traders sold ahead of portfolio insurers.

The central result is counter-intuitive. One might expect rivals of a distressed trader to provide liquidity by buying its shares cheaply. Instead they first *sell* with it, pushing the price below its new long-run level, and then buy back at the bottom. The distressed trader's extra loss is exactly the predators' profit.

This report does three things.

1. **Replication** (Section 3). We build an exact discrete-time simulator of the model and reproduce BP's Propositions 1–9 and Figures 1–3 to the paper's printed precision.

2. **Verification** (Section 4). BP derive the equilibria analytically. We check them numerically and independently: best responses are linear programs, and none improves on BP's strategies. We also show that traders who start without any predation learn it on their own.

3. **Extensions** (Section 5). Using the same tools we study three questions the paper raises but does not quantify: how much worse collusion makes predation and whether it is stable; how contagion spreads when survival hurdles are computed rather than assumed; and what drives overshooting when predators hold unequal positions.

2 The model

2.1 Setting

There is a riskless asset with zero interest rate and a risky asset with supply S and expected final value μ at time T. Long-term investors have a downward-sloping demand curve, so if I large strategic traders hold $X(t) = \sum_i x_i(t)$ shares in total, the market-clearing price is

$$p(t) = \mu - \lambda \big(S - X(t)\big), \tag{1}$$

where λ measures illiquidity. Each strategic trader chooses a trading intensity $a_i(t)$, so $x_i(t) = x_i(0) + \int_0^t a_i \, ds$, subject to a position limit $|x_i| \le \bar{x}$ with $I\bar{x} < S$ (limited capital).

Trading is not instantaneous. Strategic traders together can trade at most A shares per unit time without a temporary impact cost; orders beyond that cost γ per share (BP eq. 8). Orders are executed with equal priority: trader i's fastest buying rate $\bar{a}$ solves $\bar{a} + \sum_{j \neq i} \min(a_j, \bar{a}) = A$ (BP eq. 9), and its fastest selling rate $\underline{a}$ solves $\underline{a} + \sum_{j \neq i} \max(a_j, \underline{a}) = -A$. We use $-A$ in the selling condition, consistent with the symmetric limit $|\sum_i a_i| \le A$ of BP eq. (7).

At time t_0 a set of I_d traders becomes distressed and must sell at least at speed A/I until it holds nothing. The remaining $I_p = I - I_d$ traders are the *predators*. Each strategic trader maximises expected final wealth,

$$\max_{a_i} \; \mathbb{E}\Big[x_i(T)\, v - \int_0^T \big(a_i(t)\, p(t) + G\big)\, dt\Big]. \tag{2}$$

2.2 The key insight: Lemma 1

Substituting (1) into (2) and integrating the trader's own term, $\int a_i x_i \, dt = \frac{1}{2}[x_i^2]_0^T$, BP show that a predator's problem reduces to

$$\min_{a_i} \int_0^T a_i(t)\, X_{-i}(t)\, dt \quad \text{subject to } x_i(T) = \bar{x}, \tag{3}$$

where X_{-i} is the holdings of everyone else. A trader's own price impact cancels out: it pays the impact when buying and earns it back when selling. Profit comes only from how *other* traders move the price. A predator therefore sells while X_{-i} is falling and buys once it stops. This is what separates predation from manipulation. For our purposes the key property is that (3) is *linear* in the predator's trades.

2.3 Discrete-time implementation

We discretise $[t_0, T]$ into steps of length dt (default 0.001) and let each trader's rate be constant within a step. Holdings are then linear within a step, and so is the price by (1). The cash spent in step k is therefore exactly

$$a_k \, dt \, \frac{p_k + p_{k+1}}{2}, \tag{4}$$

so the simulated wealth equals the continuous-time integral with no discretisation error whenever strategy switches fall on the grid. When a switch falls inside a step, the step receives the strategy's average rate over it, which keeps holdings and prices exact at every grid point. Exactness lets our tests compare against BP's numbers to many decimal places rather than with loose tolerances. Trading faster than the equal-priority caps is charged $\gamma = \lambda I \bar{x}$ per excess share, the smallest value BP allow; BP's equilibria never pay it.

Throughout, the base parameters are BP's: $S = 40$, $\mu = 140$, $\lambda = 1$, $A = 20$, $t_0 = 5$, $T = 7$.

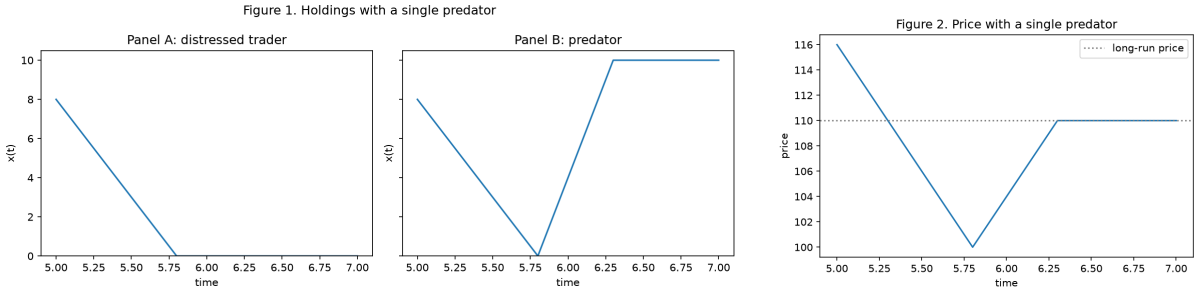


Figure 1: Proposition 1. Left: holdings of the distressed trader and the predator (BP Fig. 1). Right: the price falls from 116 to 100 and recovers to 110 (BP Fig. 2).

Table 1: Proposition 1: our simulation against BP's printed values.

Quantity	BP	Simulation
Price at $t = 5$, 5.8, 6.3, 7	116, 100, 110, 110	116, 100, 110, 110
Predator's holding at $t = 5.8$ and 6.3	0 and 10	0 and 10
Gain from selling the last share (vs. selling one less)	$101.00 - 100.50 = 0.50$	0.50
Paper / orderly / distressed value of the position	ordering only	928 / 896 / 864

3 Replication

3.1 A single predator (Proposition 1)

With $I = 2$, one distressed trader and one predator, each holding $x(t_0) = 8$ with capacity $\bar{x} = 10$, both sell at $A/2 = 10$ shares per unit time until the distressed trader is out at $t = 5.8$. The predator then buys back at speed A up to $\bar{x}$. Figure 1 shows our simulation; Table 1 compares it with BP.

The price overshoots its new level of 110 by 10. The striking feature is that the predator keeps selling even below 110. BP's argument, which our simulation reproduces exactly: the last share sells at an average of 101.00 and is bought back at an average of 100.50, a profit of 0.50 over a strategy that sells one share less.

3.2 Multiple predators (Propositions 2, 2′ and 3)

With several predators a second force appears: each wants to rebuild its position before the others. In equilibrium (Proposition 2), when $x(t_0) \geq \frac{I_p - 1}{I - 1}\,\bar{x}$, predators sell at A/I for

$$\tau = \frac{x(t_0) - \frac{I_p - 1}{I - 1}\,\bar{x}}{A/I},$$

then buy at $AI_d/(I(I_p - 1))$, a speed that keeps each predator's X_{-i} flat, until the distressed traders are out. Figure 2 reproduces BP's Figure 3, with a third of traders distressed, total holdings $I\,x(t_0) = 16$ and total capacity $I\bar{x} = 20$. The simulated price equals BP's closed form (eq. 20) at every grid point for $I = 3$, 9 and 27.

Table 2: Overshoot with multiple predators (long-run price 113.33 in every case).

I	Predators I_p	Lowest price	Overshoot
3	2	110.00	3.33
9	6	112.50	0.83
27	18	113.08	0.26

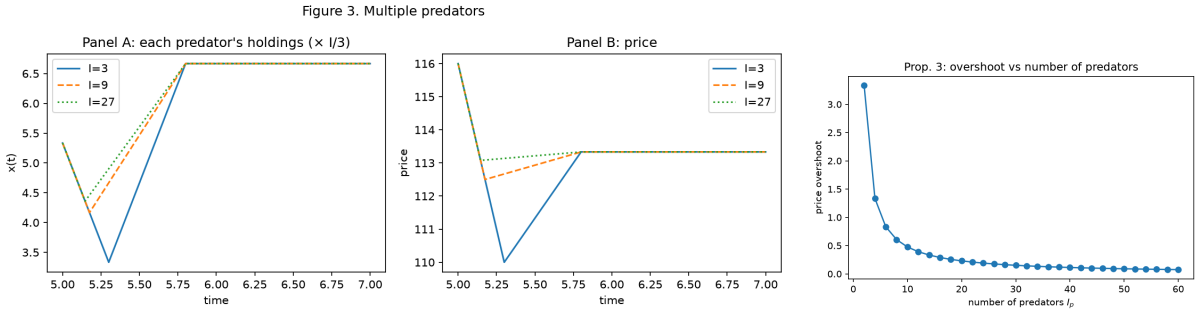


Figure 2: Left: holdings and price with $I = 3, 9, 27$ (BP Fig. 3). Right: Proposition 3, the overshoot falls toward zero as predators are added with the total holdings fixed.

As Proposition 3 states, the overshoot is positive, falls strictly with the number of predators, and vanishes in the limit: we find it strictly decreasing for $I = 3, \ldots, 147$ and below 10^{-3} at 20,000 predators (Table 2).

When predators have plenty of spare capacity, $x(t_0) < \frac{I_p - 1}{I - 1}\,\bar{x}$, competition to buy dominates and there is no predation (Proposition 2′). Predators first buy at the market's full speed, $A(I + I_d)/(I_p I)$ each, until each X_{-i} reaches its final level $(I_p - 1)\bar{x}$, which takes

$$\tau' = \frac{(I_p - 1)\bar{x} - (I - 1)\,x(t_0)}{A\left(1 - (I + I_d)/(I_p I)\right)},$$

and then absorb exactly the distressed selling. With $I = 3$, $x(t_0) = 4$, $\bar{x} = 10$, the price rises steadily from 112 to 120 with no overshoot. These dynamics apply when the fast-buying phase ends before the distressed traders are out, $t_0 + \tau' \le t_0 + x(t_0)I/A$ (in this example, $x(t_0) \ge 3.33$); our code checks this condition.

3.3 Valuation and disclosure (Propositions 5–7)

BP distinguish three values of a position: the *paper* (mark-to-market) value $x\,p$, the *orderly* liquidation value $x(p - \lambda x/2)$ from a secret sale, and the *distressed* liquidation value raised while being preyed upon. Proposition 5 orders them as paper > orderly > distressed; we confirm this for BP's example ($928 > 896 > 864$) and the Figure 3 settings. Proposition 6 says more predators raise the distressed value but never to the orderly value. With $I\,x(t_0) = 18$, the total distressed value rises from 676 with two predators towards 683.6 with 200, still below the orderly value of 690 (Figure 3, left).[1]

Proposition 7 endogenises positions before t_0: with probability π a random trader becomes distressed. Traders hold $(1 - \pi/I)\bar{x}$ if positions are not disclosed and $(1 - \pi/(I - 1))\bar{x}$ if they are. Disclosure makes traders hold less (for $I = 5$, $\pi = 0.5$: 8.75 instead of 9.0), but never little enough to prevent predation (Figure 3, right). This is BP's explanation for hedge-fund secrecy.

3.4 Market design: front-running and batch auctions (Propositions 8–9)

If the distressed traders only start selling at a later time t_1, predators front-run: they sell first at A/I_p each, wait, and buy back while the distressed traders sell (Proposition 8). If instead trading is halted and all orders clear in a batch auction at t_0, as with circuit breakers, predators cannot walk down the demand curve and buy part of their shares in the auction (Proposition 9). Our simulated prices equal BP's closed forms (eqs. 25–26) at every grid point. Proposition 8's strategy requires each predator's position during front-running, $x(t_0) - A\tau/I_p$, to stay within $-\bar{x}$; this holds in BP's examples and our code checks it.

[1] Proposition 6 requires $x(t_0) \ge \bar{x}\sqrt{I_p/I}$, which the Figure 3 setting ($I\,x(t_0) = 16$) does not meet, so we use $I\,x(t_0) = 18$.

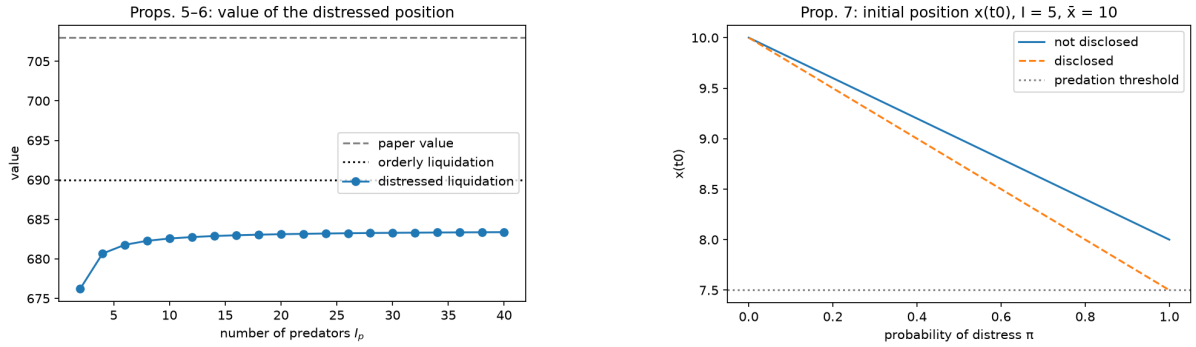
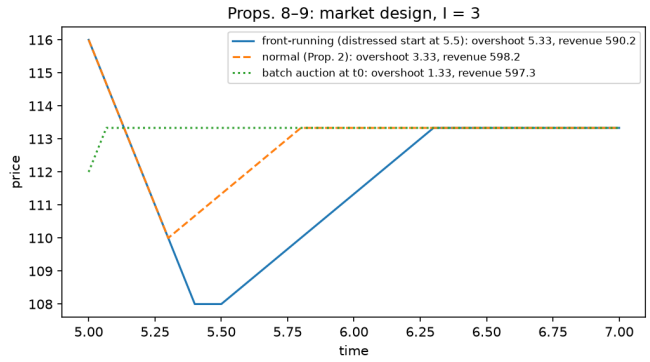


Figure 3: Left: Propositions 5–6, value of the distressed position against the number of predators. Right: Proposition 7, initial positions with and without disclosure, against the predation threshold.



Market	Overshoot	Revenue
Front-running	5.33	590.2
Normal (Prop. 2)	3.33	598.2
Batch auction	1.33	597.3

Figure 4: Propositions 8–9 with $I = 3$ (one distressed trader, front-running until $t_1 = 5.5$). Revenue is the distressed trader's total proceeds.

Figure 4 compares the three market designs. Front-running deepens the overshoot (the price falls to 108) and lowers the distressed trader's revenue, while the batch auction cuts the overshoot from 3.33 to 1.33. We add one observation: the batch auction stabilises the price but does *not* raise the distressed seller's revenue. In continuous trading the seller's first shares fetch prices near 116; in the auction all shares sell at the single clearing price of 112. With one predator the two effects cancel exactly (864.0 in both cases); in every multi-predator setting we tried the auction was slightly worse for the seller (e.g. 597.3 against 598.2 here, and 636.0 against 654.0 with two distressed traders). Circuit breakers calm prices, but their benefit to the trader in distress is not automatic.

4 Verification

4.1 Best responses as linear programs

BP derive their equilibria analytically. To check them independently we compute each predator's best response numerically. Fix every other trader's strategy; then X_{-i} is fixed, and by Lemma 1 trader i's payoff on our grid is a constant minus

$$\lambda \sum_k a_k \, dt \, \frac{X_{-i,k} + X_{-i,k+1}}{2},$$

which is linear in the trader's trades. The best response is therefore a linear program, and its optimum is exact for the discretised game. We take the trader's holdings $x_0, \ldots, x_N$ as variables, so the position limit becomes simple bounds, the speed caps become a sparse two-band constraint

Table 3: Payoff of the last predator: BP's strategy against the best possible response, everyone else held at BP's strategy.

Case	BP's strategy	Best response (LP)
Proposition 1	1214.000000	1214.000000
Proposition 2, $I = 3$	787.111111	787.111111
Proposition 2, $I = 9$	261.444444	261.444444
Proposition 2′, $I = 3$, $x(t_0) = 4$	702.000000	702.000000
Proposition 8, $I = 3$, $t_1 = 5.5$	791.111111	791.111111

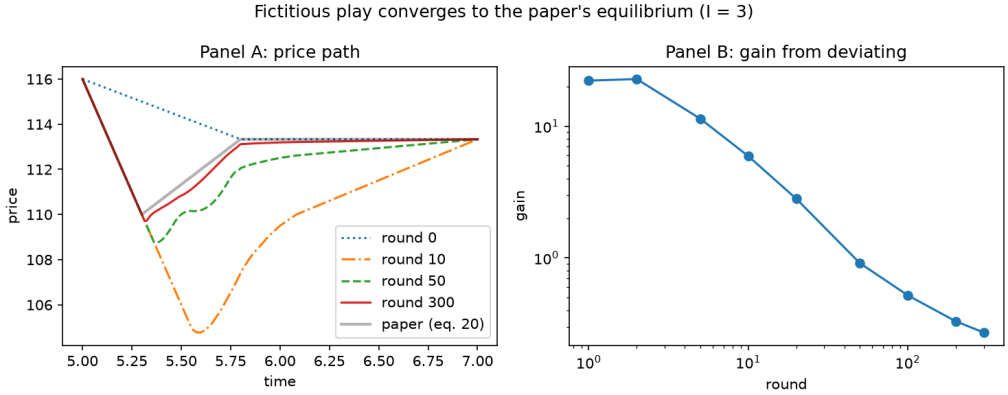


Figure 5: Fictitious play from a naive start ($I = 3$). Left: price paths. Predators first learn to prey and overdo it (round 10), then converge to BP's path. Right: the most any predator could gain by deviating, on log scales.

on $x_{k+1} - x_k$, and $x_0 = x(t_0)$, $x_N = \bar{x}$ are fixed. With 2,000 steps each LP solves in 10–30 ms with HiGHS [7].

Table 3 shows that no predator can improve on BP's strategy: the best response pays exactly the same, to six decimal places, in every case (our tests check every predator, not only the last). The best response need not coincide with BP's strategy, since BP note that the buy-back speed is free (their footnote 13); but the tests confirm the part that is pinned down, such as the predator's position when the selling stops. Two further checks show the comparison is meaningful. A predator that simply buys from t_0 gains more than 1 by switching to the LP's strategy. And if the speed caps are removed, the LP earns 1324 instead of 1214, so the check would detect a wrong speed limit. Finally, solving the LP for every end position from $-\bar{x}$ to $\bar{x}$ confirms Lemma 1's premise that ending at $\bar{x}$ is optimal: the best payoff rises all the way to $\bar{x}$ (from 414 to 1214 in Proposition 1).

4.2 Learning the equilibrium

Is the predatory equilibrium only a fixed point, or would traders find it? We start the predators on a naive strategy, simply buying steadily while the distressed traders sell, and let them respond to each other.

Plain best-response iteration cycles: the overshoot jumps between 12 and 26 for twelve rounds without settling ($I = 3$ and 9), because best responses are all-or-nothing (against an exploitable opponent a predator dumps its position at full speed). We therefore use fictitious play [4, 5]: in round k each predator moves a step $1/(k + 2)$ toward its best response, i.e. it plays the average of its past best responses. Because BP's Proposition 2 is the unique *symmetric* equilibrium, we let all predators share one strategy (one LP per round).

Figure 5 and Table 4 show the result. At round 0 there is no predation. By round 10 the predators have learned to prey but overdo it, and the price falls to about 105, far below BP's

Table 4: Fictitious play: largest distance from BP's price path (eq. 20) and gain from deviating.

	$I = 3$		$I = 9$	
Round	Price gap	Gain	Price gap	Gain
1	13.33	22.22	13.33	65.81
10	7.27	5.93	14.51	19.95
50	1.89	0.91	4.36	3.25
200	0.61	0.33	1.66	0.94
1000	0.24	0.13	0.53	0.30

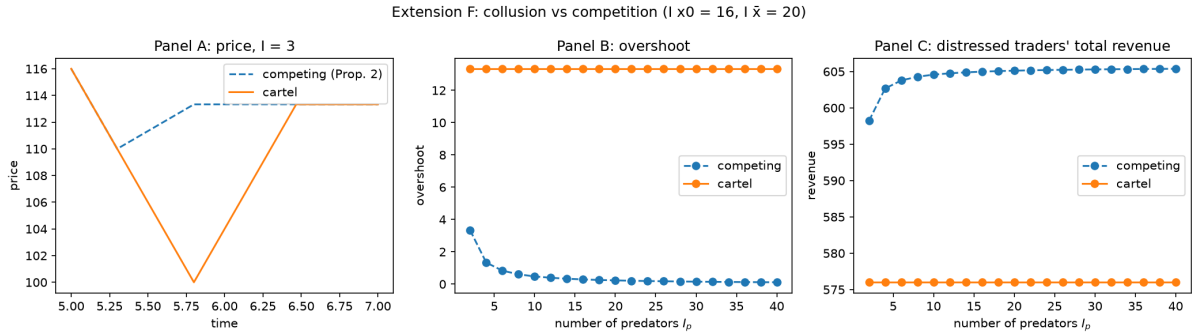


Figure 6: Collusion against competition with BP's Figure 3 totals. A cartel keeps an overshoot of 13.33 whatever its size, while competition erodes it.

110. From there the path tightens onto BP's, and the gain from deviating falls steadily (roughly like $1/k$, typical of fictitious play). When each predator instead keeps its own strategy and starts from a different one, both still converge to the symmetric equilibrium (positions at $t_0 + \tau$ of 3.335 and 3.323 after 300 rounds, against 3.333 each), consistent with BP's uniqueness result. *Predation is not only an equilibrium: traders who merely respond to each other discover it.*

Averaging strategies can make the market as a whole trade faster than A: with six predators in round 1, each adopts a response that was feasible only if the others stayed put. This is why the simulator charges the impact cost γ for excess trading (Section 2.3); the figures above include it. Since a trader can exceed its cap only if the whole market trades faster than A, the caps need computing only at such steps, so equilibrium simulations pay nothing for this.

5 Extensions

5.1 Collusion

BP note that colluding predators would "jointly act like a single predator", selling until the distressed traders are out and only then rebuilding, and that tacit collusion cannot arise in their model. We build the cartel in two independent ways and check that they agree. In closed form, each predator sells at A/I alongside the distressed traders, then all rebuild at A/I_p each. Numerically, Lemma 1's algebra applies to the group's total holdings, so the cartel solves one linear program for a single trader with capacity $I_p\bar{x}$ facing only the distressed traders. The two match to 10^{-6}.

Figure 6 and Table 5 give four results.

- **The cartel's overshoot does not shrink with its size.** It always sells down to $\mu - \lambda S = 100$, so the overshoot is $\lambda I_p \bar{x} = 13.33$ however many members it has, whereas competition drives the overshoot toward zero.

- **Collusion is a pure transfer.** The predators gain exactly what the distressed traders

Table 5: Competing predators against a cartel.

I	Overshoot	Distressed revenue	Predators' total payoff	Gain from cheating
3	$3.33 \to 13.33$	$598.22 \to 576.00$	$1574.22 \to 1596.44$	16.67 (2.1%)
9	$0.83 \to 13.33$	$603.78 \to 576.00$	$1568.67 \to 1596.44$	11.34 (4.3%)
27	$0.26 \to 13.33$	$605.06 \to 576.00$	$1567.38 \to 1596.44$	—

lose (22.22 with $I = 3$, 27.78 with $I = 9$); long-term investors are unaffected, as BP remark.

- **Collusion is unstable.** A member gains by rebuilding early, before the others, confirming numerically that collusion is not an equilibrium. The temptation is a larger share of a member's profit in a larger cartel (2.1% with two members, 4.3% with six), the usual reason large cartels are hard to sustain.
- **A cartel preys even where competition would provide liquidity.** With spare capacity ($I = 3$, $x(t_0) = 4$, $\bar{x} = 10$), competing predators buy at once and there is no overshoot (Proposition 2′), but a cartel still sells first and the price goes $112 \to 100 \to 120$, an overshoot of 12.

Repeated interaction, in which a cartel could punish cheaters and sustain tacit collusion, is outside the model; Carlin, Lobo and Viswanathan [2] study it.

5.2 Contagion

BP's Section III.B endogenises distress. A trader must liquidate if its wealth falls to a threshold $\underline{W}$. Its survival hurdle $W(I_d)$ is the highest wealth at t_0 at which it *cannot* avoid distress if I_d traders, itself included, are expected to fail (BP eq. 21), and Proposition 4 states that $W(I_d)$ increases with I_d. BP illustrate the resulting contagion with a stylised figure but do not compute the hurdles.

Definition. We make eq. (21) concrete. The other traders are $I_d - 1$ distressed traders and I_p predators playing the I_d-distressed equilibrium. Wealth is mark-to-market, $W_i = x_i\, p + \text{cash} + \text{other assets}$, and on our grid its change over a step is exactly $\lambda[\frac{1}{2}(x_{k+1}^2 - x_k^2) + \bar{x}_k\, \Delta X_{-i,k}]$, with $\bar{x}_k$ the average holding over the step. Then

$$W(I_d) = \underline{W} + D(I_d), \qquad D(I_d) = -\max_{a_i} \min_{t} \Delta W_i(t),$$

the smallest worst-case drop trader i can achieve with *any* strategy.

Computation. The x^2 term makes this max–min problem non-convex. We solve it by sequential linear programming, a convex–concave procedure [6]: replace $\frac{1}{2}x^2$ by its tangent at the current path (never above it, so every LP solution is truly achievable), solve the LP, re-linearise at the new path and repeat. Each step can only improve, and we run it from several starting paths. In experiments, 25 random starting paths per I_d all reached the same values, and the values are identical at $dt = 0.005$, 0.0025 and 0.001.

Results. We use six traders A–F as in BP's Figure 4 ($\bar{x} = 6$, $x(t_0) = 5$, $T = 9$, other parameters as before).

Table 6 confirms Proposition 4 under both rules: the more failures expected, the fiercer the predation and the higher the hurdle (strictly up to $I_d = 5$; from 5 to 6 it is flat, because a lone predator sells alongside the distressed traders just as a sixth distressed trader would). The

Table 6: Survival hurdles above the threshold, $D(I_d) = W(I_d) - \underline{W}$.

I_d	1	2	3	4	5	6
Any strategy (eq. 21)	0	27	48	63	72	72
Selling or holding only	5	34.5	57	70.5	75	75

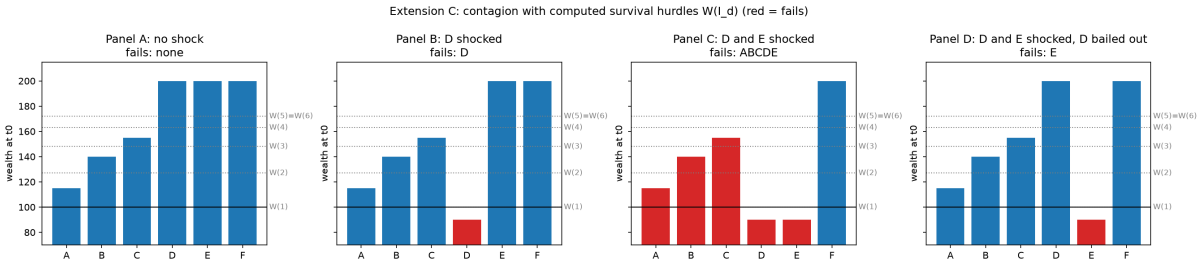


Figure 7: Contagion with the computed hurdles ($\underline{W} = 100$, so $W(1), \ldots, W(6) = 100, 127, 148, 163, 172, 172$). Red bars fail.

optimal survival strategy is notable: **sell at the speed limit for a while, then buy hard as the price bottoms**. The trader's own buying props up the price, and with it the paper value of its holdings, exactly when its wealth would otherwise be lowest. This mirrors the distressed trader who buys to support the price in Attari, Mello and Ruckes [3]. It matters: with $I_d = 4$ the best strategy limits the drop to 63, against 95 for holding and 71 for selling out. We therefore report both the literal eq. (21) hurdles and a variant in which a trader in trouble may only sell or hold.

An outcome with I_d failures is consistent if exactly the I_d traders with wealth at most $W(I_d)$ fail. The *ripple* starts from the traders forced out and adds those who fall below the hurdle for the current number of failures. With wealth levels placed as in BP's Figure 4 (A, B and C between consecutive hurdles; D, E and F safe), Figure 7 shows four scenarios.

- **No shock:** the only consistent outcome is no failure.
- **D shocked:** 1, 2, 3 or 4 failures are all consistent. If the market expects four failures, predation is fierce enough to cause them: a panic. Without panic only D fails.
- **D and E shocked:** the unique outcome is the ripple D, E $\to$ A $\to$ B $\to$ C, exactly as BP describe.
- **D and E shocked, D bailed out:** only E fails. Recapitalising one trader stops the cascade, BP's argument for central-bank intervention.

5.3 Predators with unequal positions

BP's footnote 15 describes, in words, the equilibrium when predators start with different positions: everyone sells at A/I until the X_{-i} of the predator with the *smallest* position reaches its final level $(I_p - 1)\bar{x}$, at a time s; then predators buy at $I_d A/(I(m-1))$ each, where m is the number still buying, and speed up as others finish. For the last remaining buyer we take the natural reading that it waits until the distressed traders are out and then buys at the full speed A. We build this strategy step by step, and the LP solver confirms it is an equilibrium (no profitable deviation, to 10^{-6}) for two predators at $x(t_0) \pm 0.5$ and ± 1, and for three predators at $x(t_0) - \delta, x(t_0), x(t_0) + \delta$ with $\delta = 0.4$ and 0.8, which exercises the speed-up rule. With equal positions it reduces exactly to Proposition 2.

BP state that, for fixed total holdings, overshooting increases with the dispersion of predators' initial positions. The mechanism yields a sharper statement. The price troughs at s, and

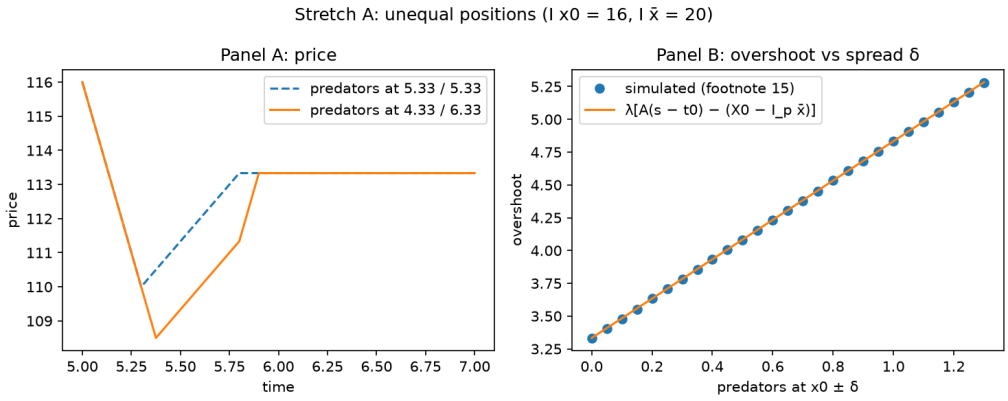


Figure 8: Unequal predator positions, $I = 3$. Left: a smaller predator prolongs the selling and deepens the trough. Right: the simulated overshoot lies exactly on equation (5).

s depends on the positions only through the smallest predator's, $x_{\min}$, and the total X_0:

$$\text{overshoot} = \lambda\big[A(s - t_0) - (X_0 - I_p\bar{x})\big], \qquad s - t_0 = \frac{\big(X_0 - x_{\min} - (I_p - 1)\bar{x}\big)\, I}{(I - 1)\, A}. \tag{5}$$

The overshoot rises by $\lambda I/(I-1)$ for each unit the smallest position sits below the rest, and the rest of the dispersion is irrelevant: **only the smallest predator matters**. It has the most of everyone else's selling still ahead of it, so it keeps selling longest, and everyone keeps selling with it. With $I = 3$ the overshoot is 3.33, 4.08, 4.83 for spreads $\delta = 0, 0.5, 1$; with $I = 4$, predators at $(3.2, 4.0, 4.8)$ and at $(3.2, 4.4, 4.4)$ have different dispersion but the same smallest position and give exactly the same overshoot, $41/15 \approx 2.733$ (Figure 8).

6 Limitations and future work

- **Distressed traders follow BP's rule.** Best responses are computed for predators only; a distressed trader's obligation to sell while holding shares depends on its own position and is not a fixed linear constraint.
- **Fictitious play converges slowly**, roughly like $1/k$, so its results are reported with tolerances.
- **Contagion is one-shot.** The cascade uses wealth at t_0, following BP's logic; wealth is not re-simulated as traders fail. The hurdle computation finds a local optimum; multiple starts make a missed better strategy unlikely but this is not a proof.
- **Time-varying liquidity.** If λ rises during a crisis, a trader's own price impact no longer cancels and Lemma 1's linear structure is lost; the sequential-LP approach of the contagion section is a natural route.
- **Empirical calibration.** Estimating λ and A from intraday data, or testing the overshoot-and-recovery pattern around index rebalances or crypto liquidations, would connect the model to data.

7 Conclusion

We implemented Brunnermeier and Pedersen's model of predatory trading as an exact discrete-time simulation and reproduced its propositions and figures. Linear-programming best responses confirm that no predator can improve on the paper's strategies, and fictitious play shows that

traders starting without predation learn it. Extending the analysis, we found that a cartel sustains a large overshoot regardless of its size but is unstable; that the best way for a trader in trouble to survive is to sell and then buy at the bottom, and that with computed survival hurdles one trader's distress can ripple through the market unless a bailout stops it; and that with unequal positions the overshoot is governed by the smallest predator alone. The results support BP's policy lessons: liquidity vanishes when it is most needed, secrecy protects large traders, circuit breakers calm prices, and bailouts can halt cascades.

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A Code and reproducibility

The project is at `https://github.com/dudegladiator/predatory-trading`. It uses Python 3.13 managed with `uv`; NumPy, SciPy (HiGHS linear programming), Matplotlib and Streamlit.

File	Contents
`src/model.py`	Parameters, price, speed caps, impact cost, simulator
`src/closed_form.py`	BP's equilibria (Props. 1–9), valuations, the cartel
`src/solver.py`	LP best responses, cartel LP, fictitious play
`src/contagion.py`	Survival hurdles by sequential LP, equilibria, cascades
`src/unequal.py`	Footnote 15 with unequal positions
`app.py`	Interactive app, one page per result
`figures.py`	All figures in this report
`tests/`	115 automated tests

Commands: `uv run pytest` runs the tests (about 30 s); `uv run python figures.py` regenerates the figures; `uv run streamlit run app.py` starts the app. Every numerical claim in this report is checked by a test against BP's printed values or against an independent method (closed form against LP, simulator against wealth formula, and so on).